

# Comprehensive Analytics Report: Delhi Traffic Inefficiency & Spatial Congestion Mapping

**Project Title:** Delhi Traffic Inefficiency & Spatial Congestion Mapping

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**Tools & Technologies Used:** Python, SQLite, SQL (CTEs, Window Functions), Pandas, NumPy, SciPy, Matplotlib, Streamlit

## 1. Executive Summary

This report provides an end-to-end quantitative analysis of urban vehicular traffic flow, temporal bottlenecks, and spatial-temporal congestion dynamics across primary arterial corridors in Delhi. Utilizing a dataset of **2,976 interval observations**, this project establishes a data-driven pipeline encompassing automated data validation, SQL-based data cleaning, advanced feature engineering, statistical hypothesis testing, and interactive web dashboarding.

The primary goal is to identify root causes of road capacity throttling and formulate actionable traffic engineering and policy recommendations for urban transit authorities.

## 2. **ASK:** Business Problem & Objectives

### 2.1 Context & Problem Statement

Metropolitan corridors in Delhi face severe daily traffic gridlocks, leading to increased transit times, higher carbon emissions, and substantial economic loss. Standard traffic management relies on fixed-time signal configurations and manual intervention, which fail to adapt to real-time fluctuations in vehicle composition and volume surges.

### 2.2 Core Research & Business Questions

1. **BQ1 (Temporal Peak Identification):** At what specific hours do arterial bottlenecks peak, and how severe is the congestion relative to daily baseline capacity?
2. **BQ2 (Vehicle Fleet Distribution):** What is the spatial road occupancy breakdown among private cars, two-wheelers, buses, and commercial heavy trucks?
3. **BQ3 (Day Part Demand Dynamics):** How does traffic volume and congestion severity vary across distinct day parts (Morning, Afternoon, Evening, Night)?

4. **BQ4 (Freight Impact):** How significantly does daytime commercial heavy truck movement impact passenger vehicle throughput and general traffic speed?

### 3. Technical Methodology & Pipeline Architecture

The analysis was executed through a modular, 17-stage analytical engineering pipeline:

Raw Data Ingestion → Data Integrity Audit → SQL ETL Cleaning Engine → Advanced SQL Analytics

Interactive Streamlit Dashboard Strategic Recommendations Statistical Tests (SciPy)  
Feature Engineering

1. **PREPARE: Database Schema & Data Ingestion:** Integrated SQLite database (`delhi_traffic.db`) with production-grade logging (`logs/etl_pipeline.log`).
2. **Automated Quality Gate:** Implemented a 5-check automated audit verifying null values, schema types, negative values, duplicates, and range boundaries.
3. **PROCESS: SQL Data Cleaning:** Utilized standard SQL transformations including `TRIM()`, `CAST()`, `COALESCE()`, and `CASE` statements for congestion banding.
4. **Advanced SQL Data Marts:** Employed Common Table Expressions (CTEs), Subqueries, and Window Functions (`DENSE_RANK()`, 1-Hour Rolling Averages) to calculate daily summaries.
5. **Feature Engineering:**
  - **Cyclical Time Encoding:** Engineered `Hour_Sin = sin(2*pi.Hour / 24)` and `Hour_Cos = cos(2*pi.Hour / 24)`.
  - **Ratios & Indicators:** Developed `Bus_To_Car_Ratio`, `Heavy_Vehicle_PCU_Share`, `Congestion_Index`, and `PCU_Lag_15Min`.

### 4. ANALYZE: Key Analytical Findings & Exploratory Data Analysis (EDA)

#### 4.1 Vehicle Composition Breakdown

- **Private Passenger Cars:** Occupy **60.1%** of total vehicular traffic volume.
- **Buses (Public Transit):** Account for **13.4%** of total volume.
- **Heavy Freight Trucks:** Represent **13.4%** of total volume.
- **Two-Wheelers (Bikes):** Represent **13.1%** of total volume.

#### 4.2 Congestion Index by Day Part

- **Evening Rush (16:00 – 19:00):** Highest average Congestion Index at **1.20** (20% above daily mean baseline), with PCU scores peaking at **~246**.
- **Morning Rush (07:00 – 10:00):** Congestion Index of **1.18** (18% above baseline), with PCU scores reaching **~233**.
- **Afternoon Window (12:00 – 16:00):** Moderate-high index of **1.14** due to commercial logistics and inter-city movement.
- **Night Window (22:00 – 05:00):** Significant load drop to an index of **0.58** (42% below daily baseline capacity).

## 5. Statistical Hypothesis Testing & Validation

To ensure all analytical findings were statistically significant and not artifacts of sample noise, rigorous parametric tests were performed using **SciPy**:

| Hypothesis Test                      | Method                    | Test Statistic | p-Value  | Statistical Decision & Interpretation                                                                               |
|--------------------------------------|---------------------------|----------------|----------|---------------------------------------------------------------------------------------------------------------------|
| Peak vs. Off-Peak PCU Difference     | Two-Sample Welch's T-Test | T=Significant  | p<0.0001 | Reject H0: Peak hours mean PCU (~233-246) is significantly higher than off-peak hours (~160).                       |
| Congestion Variance Across Day Parts | One-Way ANOVA             | F=Significant  | p<0.0001 | Reject H0: Statistically significant variance in congestion severity across Morning, Afternoon, Evening, and Night. |
| Bus Count vs. Total PCU Score        | Pearson Correlation (r)   | r=0.871        | p<0.0001 | Strong positive linear correlation; buses contribute high relative spatial occupancy per unit.                      |
| Truck Count vs. Passenger Car Count  | Pearson Correlation (r)   | r=-0.628       | p<0.0001 | Strong inverse correlation; daytime heavy truck traffic chokes passenger vehicle throughput.                        |

## 6. Business Insights Summary

1. **Private Car Inefficiency:** Private cars dominate **60.1%** of total volume, causing significant Single-Occupancy Vehicle (SOV) congestion on key corridors.
2. **Bus Queue Bottlenecks (\$r = 0.871\$):** Buses strongly correlate with high PCU scores due to frequent curb-side stopping patterns that interrupt continuous traffic flow in general lanes.
3. **Freight Choking Effect (\$r = -0.628\$):** Mixing heavy commercial trucks with commuter passenger traffic during daytime hours drastically reduces average corridor speeds and usable lane capacity.

## 7. SHARE: Dashboard & Deployment

A live, interactive web dashboard was deployed using **Streamlit** (`src/dashboard_app.py`), enabling urban planners and stakeholders to:

- Filter traffic metrics dynamically by Day Part and Peak/Off-Peak windows.
- Monitor real-time KPI cards for Total Volume, Mean PCU, Car Volume Share, and Peak Bottleneck PCU.
- Inspect hourly congestion line charts, vehicle composition donut charts, and granular data marts.

## 8. ACT: Actionable Policy & Engineering Recommendations

## 8.1 Short-Term Operational Adjustments (0 – 3 Months)

- **Dynamic Adaptive Traffic Signals:** Replace static timer signals with adaptive signal controllers programmed to extend green light timing by **25–30%** on main arterial directions during the Evening Peak window (16:00–19:00).
- **Peak-Hour Heavy Freight Restrictions:** Enforce strict entry bans for commercial multi-axle trucks on major corridors during peak hours (**07:00–10:00 AM** and **04:00–09:00 PM**).

## 8.2 Medium-Term Infrastructure Enhancements (3 – 12 Months)

- **Dedicated Bus Rapid Transit (BRT) Corridors:** Demarcate priority bus lanes on high-density corridors to isolate frequent stopping patterns from high-speed passenger traffic.
- **High-Occupancy Vehicle (HOV) Lanes:** Reserve the extreme right lane for HOV-2+ vehicles (cars carrying 2 or more passengers) during peak rush hours to incentivize carpooling.

## 8.3 Long-Term Structural Policies (12 – 24 Months)

- **Mandatory Night Logistics Shift:** Require major commercial freight and warehouse operations to shift loading/unloading activities exclusively to the Night Window (**22:00–05:00**), leveraging the low baseline congestion index (**0.58**).
- **Automated Congestion Pricing:** Implement ANPR/RFID-based dynamic tolling during peak hours (16:00–19:00) on bottleneck corridors to reduce non-essential private vehicle travel.

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